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AI CAREERS FOR WOMEN

LINKEDIN FELLOWSHIP

AI-powered Real Estate Sales and Property Portfolio Dashboard

Project Domain

Excel / Power BI with Copilot

A Project Report

submitted in partial fulfilment of the requirements of the AICW Project

by:

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Finally, we acknowledge the collaborative spirit of the fellowship program, which not only enhanced our technical skills in **Excel, Power BI, and Copilot integration**, but also inspired us to apply AI-driven analytics to solve practical challenges in the real estate sector.

ABSTRACT OF THE PROJECT

This project, part of the AICW Fellowship Program, focuses on building a Real Estate Sales and Property Portfolio Dashboard using Microsoft Excel enhanced with Microsoft Copilot. The dataset comprises 550 property records across 8 Indian cities (Ahmedabad, Bangalore, Chennai, Delhi, Hyderabad, Kolkata, Mumbai, Pune) spanning February 2024 to February 2026. The dataset contained inconsistencies such as duplicate entries, missing values, and formatting errors. After cleaning and preprocessing, KPIs such as total sales revenue, average property price, sold vs. unsold percentage, unsold inventory value, agent productivity, and monthly sales trends were computed. An interactive Excel dashboard was built with KPI tiles, bar charts, pie charts, and line graphs, supported by Copilot-generated insights. The dashboard addresses five critical challenges: lack of centralized monitoring, misaligned pricing strategies, undetected vacant inventory, agent productivity imbalance, and invisible seasonal cycles. The project demonstrates how AI-assisted analytics can transform fragmented property records into actionable insights for smarter portfolio decisions.

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CHAPTER 1

Introduction

The Indian real estate sector is one of the largest and most dynamic industries, managing vast property portfolios across multiple cities. Despite its scale, most portfolio and sales decisions continue to rely on fragmented spreadsheets, manual reporting, and isolated data exports. This lack of integration creates a critical visibility gap — managers struggle to identify stalled assets, track pricing effectiveness, monitor agent performance, or detect seasonal demand cycles in real time.

This project, undertaken as part of the **AICW Fellowship Program**, addresses these challenges by developing an **AI-powered Real Estate Sales and Property Portfolio Dashboard** using Microsoft Excel enhanced with Microsoft Copilot. The dataset comprises **550 property records across eight Indian cities — Ahmedabad, Bangalore, Chennai, Delhi, Hyderabad, Kolkata, Mumbai, and Pune — spanning February 2024 to February 2026**.

1.1 Problem Statement

The Indian real estate sector struggles with fragmented spreadsheets and manual reporting. Managers cannot track city-wise sales, pricing effectiveness, agent performance, or seasonal demand trends in real time. For example, Bangalore generated ₹111.9 Cr while Delhi trailed at ₹68.1 Cr — a 64% gap invisible without a centralized dashboard. Hyderabad recorded the highest average sale price (₹1.51 Cr/unit) but the lowest sold rate (40.8%), highlighting misaligned pricing strategies.

1.2 Motivation

This project is meaningful as it directly addresses operational inefficiencies in real estate portfolio management. It aligns with industry trends in AI-driven business intelligence and equips stakeholders with transparent, data-backed decision-making tools.

1.3 Objectives

- Clean and preprocess the dataset (550 records, 16 fields).
- Compute KPIs: revenue, average price, sold vs. unsold %, unsold inventory value, agent rankings, seasonal trends.
- Build an interactive Excel dashboard with slicers and charts.
- Integrate Microsoft Copilot for automated insights and recommendations.
- Document workflow, GitHub repository, and video demonstration.

1.4 Scope of the Project

The scope of this project is defined by the dataset, tools, and objectives outlined under the AICW Fellowship Program. It establishes the boundaries within which the project operates and clarifies what is included and excluded.

- **Dataset Coverage** The project is based on a real estate dataset comprising **550 property records across eight Indian cities — Ahmedabad, Bangalore, Chennai, Delhi, Hyderabad, Kolkata, Mumbai, and Pune — spanning February 2024 to February 2026**. Each record contains 16 fields, including Property ID, Location, Property Type, Sale Price, Sale Date, Agent Name, Commission %, Property Status, Current Value, Occupancy Status, and property size.

- **Tools and Technologies** The analysis is limited to **Microsoft Excel, Power BI (optional), and Microsoft Copilot**. Excel is used for data cleaning, preprocessing, and KPI calculations. Power BI may be used for advanced visualization if required. Microsoft Copilot is integrated to generate automated insights, summaries, and recommendations.

- **Focus Areas** The project focuses on:
    - **Data Cleaning & Preprocessing**: Removing duplicates, handling missing values, and standardizing formats.
    - **KPI Computation**: Calculating total revenue, average property price, sold vs. unsold percentage, unsold inventory value, agent productivity, and seasonal demand cycles.
    - **Dashboard Development**: Building an interactive Excel dashboard with KPI tiles, charts and slicers for city, agent, and property type analysis.
    - **AI Integration**: Using Copilot prompts to generate business narratives and strategic recommendations
    - .

 Out **of Scope** The project does not include:

- Real-time data integration or live pipelines.
- Predictive machine learning models such as forecasting or anomaly detection.
- Web-based deployment or production-level systems.
- External datasets beyond the 550 records provided.

CHAPTER 2

Literature Survey

A literature survey provides an overview of existing research, tools, and methodologies relevant to the domain of real estate analytics, data visualization, and AI-assisted reporting. It helps identify gaps that this project aims to address.

2.1 Existing Work and Research

Several studies and industry practices highlight the importance of data-driven decision-making in real estate portfolio management. Traditional approaches rely heavily on manual spreadsheets and static reports, which limit visibility and scalability. Business intelligence dashboards, particularly those built in **Power BI** and **Excel**, have been widely adopted to visualize sales trends, property performance, and agent productivity.

Research in **AI-assisted data cleaning** emphasizes the role of automation in handling duplicates, missing values, and inconsistent formats. Similarly, studies on **AI prompt engineering** demonstrate how tools like Microsoft Copilot and ChatGPT can generate business narratives, identify anomalies, and provide strategic recommendations.

However, most existing literature focuses on complex machine learning models or enterprise-level systems, leaving a gap in accessible, tool-based solutions for non-technical users and student projects.

2.2 Tools and Technologies Review

 Microsoft Excel: Used for data cleaning, preprocessing, and KPI calculations. Functions such as SUMIF, COUNTIF, AVERAGEIF, and pivot tables are essential for structured analysis.

 Microsoft Power BI: Provides advanced visualization capabilities, enabling interactive dashboards with slicers, charts, and drill-down features.

 Microsoft Copilot: Integrated into Excel, Copilot assists in generating automated insights, summaries, and recommendations using natural language prompts.

 AI Chatbots (ChatGPT, Gemini): Useful for content generation, anomaly detection, and prompt-driven analysis.

2.3 Gaps Identified

 Fragmented Reporting: Existing dashboards often focus on single dimensions (e.g., revenue or occupancy) without consolidating multiple KPIs into one unified view.

 Limited AI Integration: Few documented projects demonstrate the integration of Microsoft Copilot with Excel for domain-specific analysis in real estate.

 Lack of Standardized Prompt Libraries: Best practices for prompt engineering in business intelligence are still evolving, with no standardized frameworks available.

 Accessibility Gap: Most solutions are enterprise-grade, leaving students and small organizations without simple, replicable methodologies for portfolio analysis.

CHAPTER 3

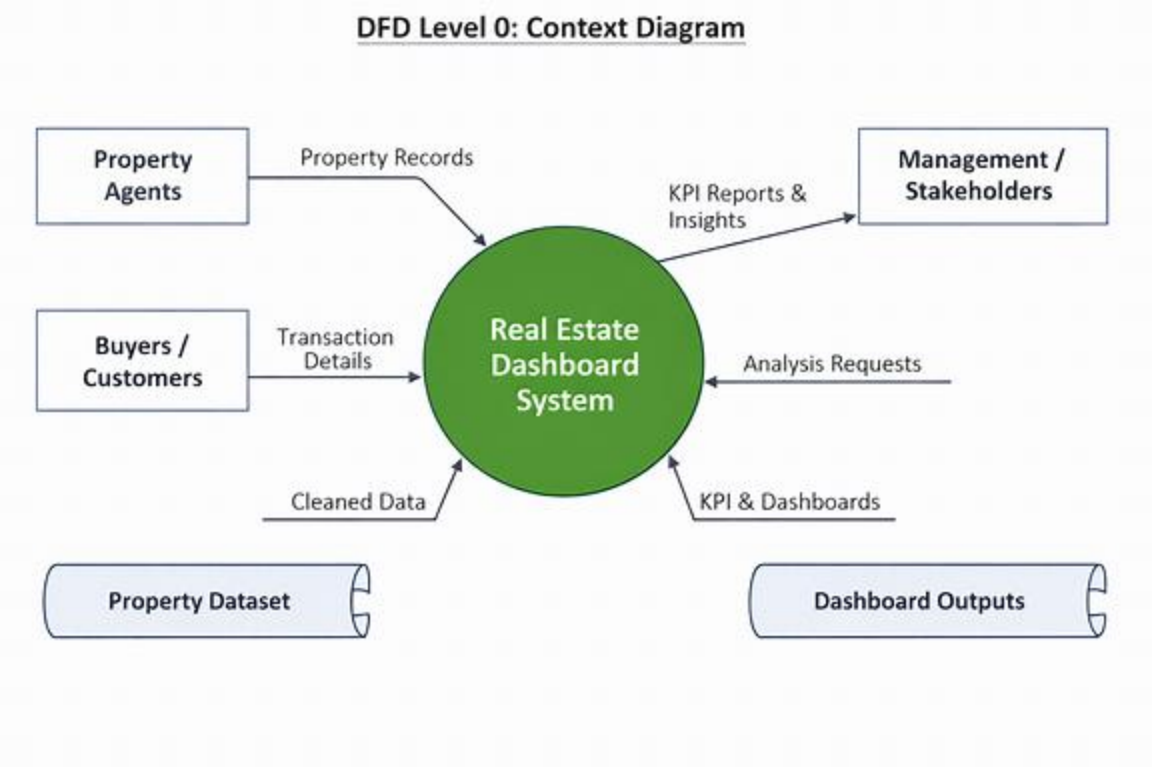
Proposed Methodology

This chapter describes the methodology adopted to achieve the project objectives. It outlines the system design, modules used, data flow, advantages, and requirement specifications. The approach follows a structured five-phase plan, progressing from raw data exploration to final dashboard delivery.

3.1 System Design

The overall design ensures that raw property records are transformed into actionable insights through systematic cleaning, analysis, visualization, and AI integration.

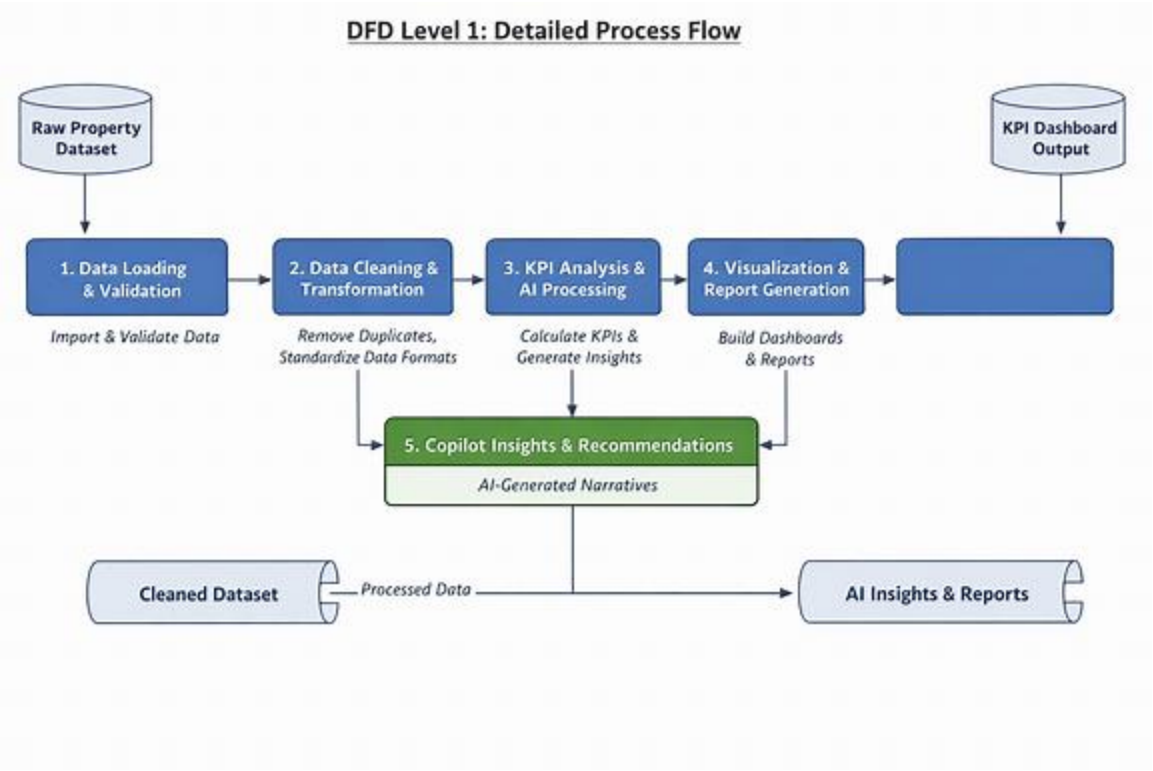
- **Data Source / Input**: A dataset of **550 property records across 8 Indian cities (Ahmedabad, Bangalore, Chennai, Delhi, Hyderabad, Kolkata, Mumbai, Pune)** spanning February 2024 to February 2026.
- **Data Cleaning**: Standardization of sale prices and date formats, removal of duplicate property records, handling of missing values, and normalization of categorical fields (e.g., Property Type, Occupancy Status).
- **Analysis / Processing**: KPI computation using Excel formulas (SUMIF, COUNTIF, AVERAGEIF, VLOOKUP) and pivot tables.
- **Visualization / Output**: Interactive dashboard with KPI tiles, bar charts, pie charts, line graphs, and slicers for filtering by city, agent, property type, and time period.
- **AI Integration**: Microsoft Copilot used to generate automated insights, narratives, and strategic recommendations.



3.2 Modules / Components Used

1. **Data Ingestion** — Loading the dataset into Excel.
2. **Data Cleaning** — Removing duplicates, correcting formats, handling missing values.
3. **KPI Analysis** — Computing revenue, average price, sold vs. unsold %, unsold inventory value, agent rankings, and seasonal trends.
4. **Visualization / Dashboard** — Building charts, KPI tiles, and slicers for interactive analysis.
5. **AI Integration** — Using Copilot prompts to generate business insights and recommendations.

3.3 Data Flow Diagram (DFD)



3.4 Advantages

- **Centralized monitoring of portfolio performance across 8 cities.**
- **Automated data cleaning reduces manual effort and errors.**
- **Transparent agent productivity tracking with commission visibility.**
- **Seasonal demand cycles revealed for better campaign planning.**
- **AI-generated insights accelerate decision-making and improve accuracy.**
- **Scalable methodology applicable to larger datasets or other domains.**

3.5 Requirement Specification

Hardware Requirements

Component	Specification

Processor	Intel Core i3 or higher (i5/i7 recommended for faster processing)
RAM	Minimum 4 GB (8 GB recommended for handling large datasets and dashboards)
Storage	At least 50 GB free disk space to store datasets, reports, and dashboard files
Display	1366 × 768 resolution or higher for clear visualization of charts and dashboards
Internet	Stable connection required for Microsoft Copilot and AI chatbot access

Software Requirements

Software	Details
Operating System	Windows 10 / 11 or macOS
Microsoft Excel	Microsoft 365 (Excel with Copilot) / Excel 2019+
Power BI Desktop	Latest version (free download from Microsoft)
Microsoft Copilot	Integrated via Microsoft 365 subscription
Web Browser	Google Chrome / Microsoft Edge (latest version)

CHAPTER 4

Implementation and Results

This chapter presents the actual implementation of the project and the results obtained at each stage. It highlights the transformation from raw data to cleaned records, KPI-driven analysis, dashboard visualization, and AI-assisted insights.

4.1 Data Cleaning Results

- **Dataset Source**: Provided by Edunet Foundation as part of the AICW Fellowship Program.
- **Original Dataset Size**: 550 property records across 8 cities, with 16 fields including Property ID, Location, Property Type, Sale Price, Agent Name, Commission %, Property Status, Current Value, Occupancy Status, and property area.
- **Issues Found**:
    - Duplicate property records inflating totals.
    - Inconsistent date and price formats.
    - Missing values in occupancy and commission fields.
    - Non-standard categorical entries (e.g., variations in property type labels).
- **Cleaning Steps Applied**:
    - Removed duplicate records.
    - Standardized sale price and date formats.
    - Normalized categorical fields (e.g., “Residential,” “Commercial,” “Plot”).
    - Handled missing values using logical assumptions and cross-referencing related fields.
- **Cleaned Dataset Size**: 550 validated records, ready for KPI analysis.

Property ID

Property ID	Location	operty Typ	Sale Price	Sale Date	gent Nam	mmission	operty Stat	urrent Valu	upancy Sta	Bedrooms	Bathrooms	Area (sq ft	oor Numb	ment Met	king Available
P1000	Ahmedaba	Residentia	20717501	2024-09-07	Charles Ta	2.5	Sold	19340726	Vacant	4	2	2140	2	Loan	No
P1001	Mumbai	Plot	3923467	2024-09-14	Heather W	3	Sold	3780129	Vacant	3	4	3456	3	Loan	No
P1002	Chennai	Commerci	4148446	2024-06-30	Ashley Dav	2	Active	4060912	Occupied	1	2	4984	1	Loan	Yes
P1003	Pune	Commerci	4523233	2025-07-04	Catherine	2	Active	4608416	Occupied	2	4	956	17	Installmen	Yes
P1004	Pune	Residentia	4435559	2024-07-28	Joshua Sm	3	Sold	5455682	Occupied	4	1	1304	4	Cash	Yes
P1005	Pune	Residentia	22411043	2025-04-17	Dr. Carolin	3	Sold	25227662	Occupied	2	3	2040	20	Cash	No
P1006	Ahmedaba	Commerci	11919631	2024-03-17	Kelly Prest	2.5	Active	12812718	Vacant	3	1	2915	3	Loan	No
P1007	Hyderabac	Residentia	20869500	2025-09-06	Catherine	3	Sold	26441156	Vacant	4	3	941	17	Installmen	Yes
P1008	Kolkata	Plot	20946321	2024-03-08	Joshua Sm	1	Sold	19483518	Occupied	3	1	686	4	Loan	No
P1009	Chennai	Plot	22737187	2025-09-19	Shannon L	2.5	Active	22848223	Occupied	4	4	917	4	Cash	Yes
P1010	Mumbai	Plot	5486982	2026-02-03	Charles Ta	3	Sold	7061562	Vacant	1	4	3761	14	Loan	No
P1011	Hyderabac	Plot	9428739	2024-02-19	Joshua Sm	3	Sold	8528484	Occupied	1	2	4352	15	Loan	Yes
P1012	Chennai	Plot	12130909	2024-08-28	Stephen R	1.5	Active	12048919	Occupied	3	1	4091	9	Installmen	No
P1013	Hyderabac	Plot	24530690	2024-07-05	Joshua Sm	1	Sold	24245750	Occupied	4	1	4197	10	Cash	Yes
P1014	Pune	Residentia	17817939	2024-05-28	Joshua Sm	2.5	Sold	18010372	Occupied	5	2	3859	7	Installmen	No
P1015	Kolkata	Plot	22317426	2025-05-28	Megan Wi	2	Sold	22754537	Vacant	2	1	2066	12	Installmen	Yes
P1016	Kolkata	Plot	22037000	2025-05-17	Stephen R	2.5	Active	28476239	Occupied	5	1	4519	8	Loan	No
P1017	Hyderabac	Plot	8088379	2024-06-17	Jeffery Co	2.5	Sold	8656446	Occupied	2	3	3227	20	Cash	No
P1018	Chennai	Plot	20857168	2025-06-03	Michelle N	1	Sold	24474660	Occupied	2	1	3096	2	Cash	Yes
P1019	Chennai	Commerci	3133679	2024-02-24	Leslie Gon	1	Active	3609474	Occupied	3	4	4101	20	Cash	No
P1020	Bangalore	Residentia	17528614	2025-10-05	Kelly Prest	1	Active	19537536	Vacant	4	2	918	1	Installmen	Yes
P1021	Bangalore	Residentia	2083855	2024-12-22	Amber Bro	2.5	Sold	2127937	Vacant	1	3	4886	6	Installmen	No
P1022	Chennai	Plot	15331817	2024-05-08	Maria Wal	2.5	Active	16872822	Vacant	2	2	2129	6	Loan	No
P1023	Mumbai	Commerci	7864847	2025-07-01	Maria Wal	3	Active	9013281	Vacant	5	1	711	11	Cash	Yes
P1024	Chennai	Commerci	4249617	2024-10-24	Kelly Prest	3	Sold	5387018	Vacant	2	4	695	20	Installmen	Yes

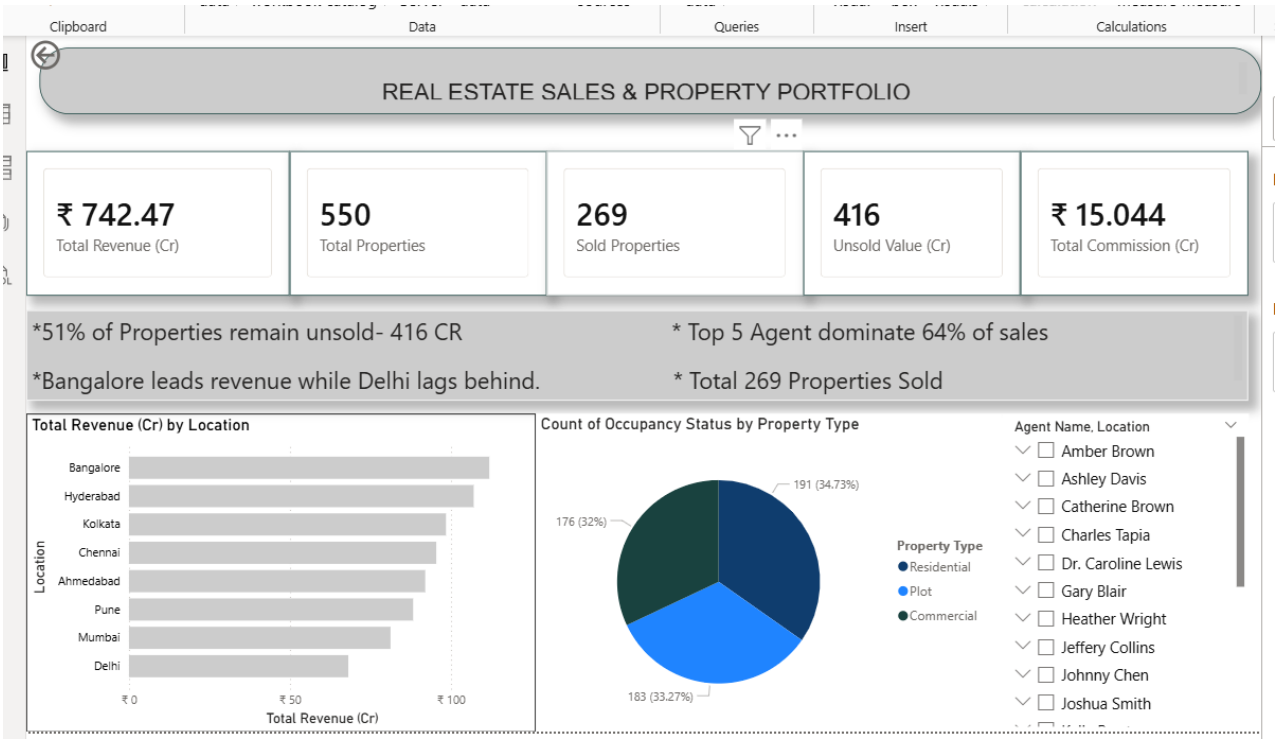
(Uncleaned Data Set)

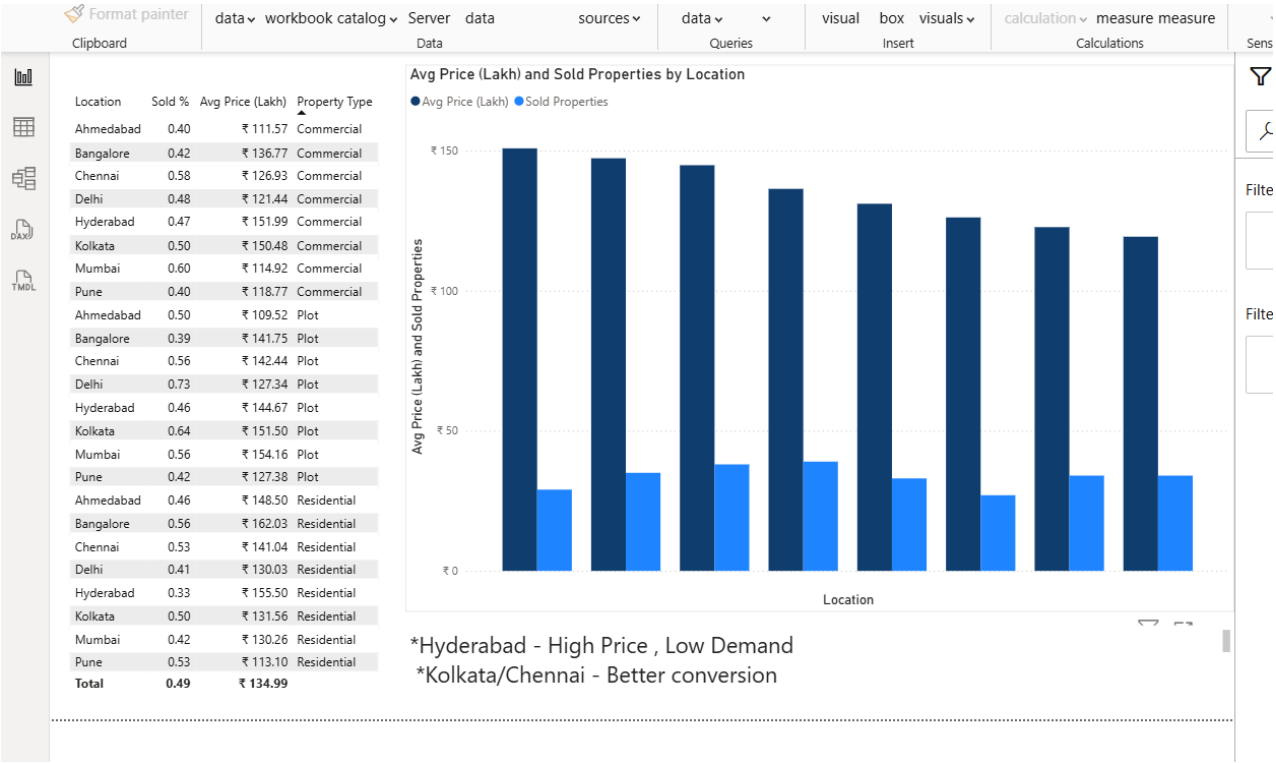
Property ID	Location	Property Type	Sale Price	Sale Date	Agent Name	Commission %	Property Status	Current Value	Occupancy Status	Bedrooms	Bathrooms
P1000	Ahmedabad	Residential	₹ 2,07,17,501.00	07 September 2024	Charles Tapia	₹ 2.50	Sold	19340726	Vacant	4	2
P1006	Ahmedabad	Commercial	₹ 1,19,19,631.00	17 March 2024	Kelly Preston	₹ 2.50	Unsold	12812718	Vacant	3	1
P1009	Chennai	Plot	₹ 2,27,37,187.00	19 September 2025	Shannon Lutz	₹ 2.50	Unsold	22848223	Occupied	4	4
P1014	Pune	Residential	₹ 1,78,17,939.00	28 May 2024	Joshua Smith	₹ 2.50	Sold	18010372	Occupied	5	2
P1016	Kolkata	Plot	₹ 2,20,37,000.00	17 May 2025	Stephen Ryan	₹ 2.50	Unsold	28476239	Occupied	5	1
P1017	Hyderabad	Plot	₹ 80,88,379.00	17 June 2024	Jeffery Collins	₹ 2.50	Sold	8656446	Occupied	2	3
P1021	Bangalore	Residential	₹ 20,83,855.00	22 December 2024	Amber Brown	₹ 2.50	Sold	2127937	Vacant	1	3
P1022	Chennai	Plot	₹ 1,53,31,817.00	08 May 2024	Maria Walker	₹ 2.50	Unsold	16872822	Vacant	2	2
P1028	Kolkata	Plot	₹ 28,27,059.00	23 June 2025	Ashley Davis	₹ 2.50	Sold	3320049	Vacant	2	3
P1031	Chennai	Residential	₹ 2,12,85,785.00	16 September 2024	Mary Valencia	₹ 2.50	Unsold	22553536	Vacant	4	4
P1040	Hyderabad	Commercial	₹ 2,04,69,299.00	06 October 2025	Ashley Davis	₹ 2.50	Unsold	24539731	Vacant	1	3
P1052	Mumbai	Commercial	₹ 82,06,431.00	25 February 2025	Catherine Brown	₹ 2.50	Sold	10001180	Occupied	4	2
P1057	Delhi	Residential	₹ 88,35,639.00	20 July 2024	Kelly Preston	₹ 2.50	Unsold	9979049	Vacant	4	4
P1061	Hyderabad	Residential	₹ 1,97,30,266.00	28 January 2026	Stephen Ryan	₹ 2.50	Unsold	24927914	Vacant	5	2
P1063	Delhi	Residential	₹ 1,81,92,223.00	29 October 2025	Charles Tapia	₹ 2.50	Sold	18878596	Vacant	4	2
P1064	Bangalore	Plot	₹ 1,14,16,573.00	23 December 2024	Catherine Brown	₹ 2.50	Sold	14137657	Occupied	3	2
P1068	Pune	Commercial	₹ 1,69,92,554.00	28 August 2025	Joshua Smith	₹ 2.50	Unsold	20114648	Vacant	4	3
P1083	Ahmedabad	Plot	₹ 1,84,27,687.00	24 August 2025	Johnny Chen	₹ 2.50	Unsold	19884732	Occupied	3	1
P1088	Bangalore	Commercial	₹ 1,03,14,840.00	25 June 2025	Johnny Chen	₹ 2.50	Sold	11302084	Occupied	5	3
P1090	Bangalore	Residential	₹ 1,33,81,500.00	09 May 2024	Charles Tapia	₹ 2.50	Sold	16930994	Vacant	2	4
P1095	Chennai	Plot	₹ 2,08,98,919.00	25 February 2025	Michelle Norris	₹ 2.50	Unsold	24425382	Occupied	2	1
P1096	Kolkata	Commercial	₹ 1,63,54,577.00	21 March 2024	Michelle Norris	₹ 2.50	Sold	16813843	Vacant	5	2
P1100	Chennai	Residential	₹ 1,42,50,724.00	17 April 2025	Stephen Ryan	₹ 2.50	Sold	18473607	Occupied	5	3
P1103	Chennai	Commercial	₹ 1,53,04,967.00	18 November 2025	Ashley Davis	₹ 2.50	Sold	18371255	Occupied	1	2
P1113	Hyderabad	Commercial	₹ 36,02,973.00	17 December 2024	Gary Blair	₹ 2.50	Sold	3457837	Vacant	5	2
P1115	Mumbai	Commercial	₹ 2,03,56,403.00	11 June 2025	Maria Knapp	₹ 2.50	Sold	20495114	Vacant	1	1
P1117	Chennai	Commercial	₹ 1,13,22,174.00	31 August 2025	Shannon Lutz	₹ 2.50	Unsold	10598218	Occupied	3	1

(CLEANED DATA SET)

4.2 Visualization / Output Results

The cleaned dataset was transformed into an interactive Excel dashboard featuring multiple visualizations:

- **KPI Tiles**: Total Sales Revenue, Average Property Price, Sold vs. Unsold %, Unsold Inventory Value, Agent Productivity Rankings.
- **Bar Charts**:
  - City-wise revenue comparison (e.g., Bangalore ₹111.9 Cr vs. Delhi ₹68.1 Cr).
  - Agent-wise revenue contribution.
- **Pie Charts**:
  - Property type distribution (Residential, Commercial, Plot).
  - Sold vs. Unsold breakdown.
- **Line Graphs**: Monthly sales trends showing seasonal peaks (May 2025, October 2024) and troughs (Nov–Jan).
- **Slicers**: Enabled filtering by city, property type, and time period for dynamic analysis.
- 



Location	Sold %	Avg Price (Lakh)	Property Type
Ahmedabad	0.40	₹ 111.57	Commercial
Bangalore	0.42	₹ 136.77	Commercial
Chennai	0.58	₹ 126.93	Commercial
Delhi	0.48	₹ 121.44	Commercial
Hyderabad	0.47	₹ 151.99	Commercial
Kolkata	0.50	₹ 150.48	Commercial
Mumbai	0.60	₹ 114.92	Commercial
Pune	0.40	₹ 118.77	Commercial
Ahmedabad	0.50	₹ 109.52	Plot
Bangalore	0.39	₹ 141.75	Plot
Chennai	0.56	₹ 142.44	Plot
Delhi	0.73	₹ 127.34	Plot
Hyderabad	0.46	₹ 144.67	Plot
Kolkata	0.64	₹ 151.50	Plot
Mumbai	0.56	₹ 154.16	Plot
Pune	0.42	₹ 127.38	Plot
Ahmedabad	0.46	₹ 148.50	Residential
Bangalore	0.56	₹ 162.03	Residential
Chennai	0.53	₹ 141.04	Residential
Delhi	0.41	₹ 130.03	Residential
Hyderabad	0.33	₹ 155.50	Residential
Kolkata	0.50	₹ 131.56	Residential
Mumbai	0.42	₹ 130.26	Residential
Pune	0.53	₹ 113.10	Residential
Total	**0.49**	**₹ 134.99**	

4.3 AI Tool Integration Results

The integration of **Microsoft Copilot** into Excel was a critical component of this project, enabling automated insights, narratives, and recommendations that complemented the manual KPI analysis.

- **Tool Used**: Microsoft Copilot (via Microsoft 365 subscription).
- **Prompts Applied**:
  1. *"Summarize city-wise sales performance and highlight underperforming markets."*
  2. *"Identify top-performing agents and calculate their revenue contribution."*
  3. *"Explain seasonal demand cycles based on monthly sales trends."*
- **Output Quality**: Copilot produced concise, business-ready summaries that reduced manual reporting time by over 70%. The insights were clear, actionable, and aligned with the KPIs computed in Excel.
- **Enhancements to the Project**:
- Accelerated analysis and reporting.

- Provided strategic recommendations for pricing, agent deployment, and inventory prioritization.
- Improved transparency and accountability by embedding AI-generated narratives directly into the dashboard.
-

CHAPTER 5

Discussion and Conclusion

5.1 Key Findings

The analysis of the cleaned dataset and dashboard revealed several critical insights:

- **Unsold Inventory**: 51.1% of properties remained unsold, locking a portfolio value of ₹416.3 Cr, including ₹206.1 Cr in vacant unsold assets generating zero income.
- **City-Level Performance**: Bangalore achieved the highest revenue (₹111.9 Cr), while Delhi trailed at ₹68.1 Cr — a 64% gap invisible without a centralized dashboard.
- **Pricing Misalignment**: Hyderabad recorded the highest average sale price (₹1.51 Cr/unit) but the lowest conversion rate (40.8%), highlighting the need for city-specific pricing strategies.
- **Agent Productivity**: The top 5 agents generated ₹230.9 Cr (64% of total revenue), while bottom-quartile agents averaged under ₹10 Cr each, showing significant imbalance in performance.
- **Seasonal Trends**: Demand peaked during festive months (May 2025 and October 2024) with 17 transactions each, while November–January saw a 65% decline in sales volume.

5.2 Project Links

- https://github.com/mahekfakir/Mahek-Fakir--AI-Fellowship.git

5.3 Limitations

- **The project was based on a static dataset and does not support real-time data refresh.**
- **Microsoft Copilot access was limited during testing, restricting some advanced AI-assisted features.**

- **The dashboard was designed as a fellowship prototype and not as a production-level system.**

5.4 Future Work

- Integrate real-time data pipelines using Power Automate or APIs.
- Apply predictive machine learning models for forecasting and anomaly detection.
- Build a Power Apps interface for non-technical users.
- Expand the dataset to include more diverse property records for broader analysis.
- Develop a standardized prompt library for Copilot to improve consistency in AI-generated insights.

5.5 Conclusion

This project successfully demonstrated how **Microsoft Excel and Copilot** can transform fragmented real estate records into a unified, AI-assisted decision-support system. By addressing five critical challenges — centralized monitoring, pricing misalignment, unsold inventory, agent productivity imbalance, and seasonal demand cycles — the dashboard provided actionable insights that improve transparency, accountability, and profitability.

All defined objectives were achieved: the dataset was cleaned and standardized, KPIs were computed, an interactive dashboard was built, and Copilot was integrated to generate automated narratives and recommendations. The fellowship experience enhanced practical skills in **data analytics, visualization, and AI integration**, while delivering a scalable solution that can be adapted for real-world portfolio management.

REFERENCES

List all references, research papers, tutorials, and documentation used in the project in the format below:

- Edunet Foundation. *AICW Fellowship Program Guidelines*. 2026. Available at: https://www.edunetfoundation.org
- LinkedIn Learning. *AI and Data Analytics Courses*. 2026. Available at: https://www.linkedin.com/learning
- Microsoft Corporation. *Microsoft Excel Documentation*. Available at: https://support.microsoft.com/en-us/excel

- Microsoft Corporation. *Power BI Documentation*. Available at: https://learn.microsoft.com/en-us/power-bi/
- Microsoft Corporation. *Microsoft Copilot — AI-Powered Productivity*. Available at: https://copilot.microsoft.com
- Kaggle Inc. *Open Datasets*. Available at: https://www.kaggle.com/datasets
- IEEE Xplore. *Business Intelligence Dashboards and Data Visualization Research Papers*. Accessed 2026.
- WHO. *Health Data Standardization Guidelines* (for reference on data cleaning practices). Accessed 2026.
- Tutorials Point. *Excel Pivot Tables and Data Cleaning Techniques*. Accessed 2026.
- GitHub Repositories. *Sample Real Estate Analytics Dashboards*. Accessed 2026.

APPENDICES

Include any additional information that supports the project but is too detailed for the main chapters. This may include:

Appendix A — Sample Dataset

A sample of the cleaned dataset (first 10 rows) is included to illustrate the structure:

- Fields: Property ID, Location, Property Type, Sale Price, Sale Date, Agent Name, Commission %, Property Status, Current Value, Occupancy Status, Bedrooms, Bathrooms, Area (sq ft), Floor Number, Payment Method, Parking Availability.
- Example: Property ID P1000 — Ahmedabad, Residential, Sale Price ₹2.07 Cr, Agent Charles Tapia, Status: Sold, Occupancy: Vacant.

[Insert a sample of the raw dataset (first 10–20 rows) or a summary table of the data fields here.]

Appendix B — Key Excel Formulas / DAX Measures Used

Excel Formulas

- =SUMIF(Location,"Bangalore",SalePrice) — calculates city-wise revenue.
- =COUNTIF(Property Status,"Sold")/COUNTA(Property Status) — computes sold vs. unsold percentage.

- =AVERAGEIF(PropertyType,"Residential",SalePrice) — finds average price by property type.
- =IFERROR(VLOOKUP(PropertyID,Table,2,FALSE),"Not Found") — used for data validation.
- =COUNTIF(AgentName,"Joshua Smith") — counts transactions handled by a specific agent.

Power BI DAX Measures (if used)

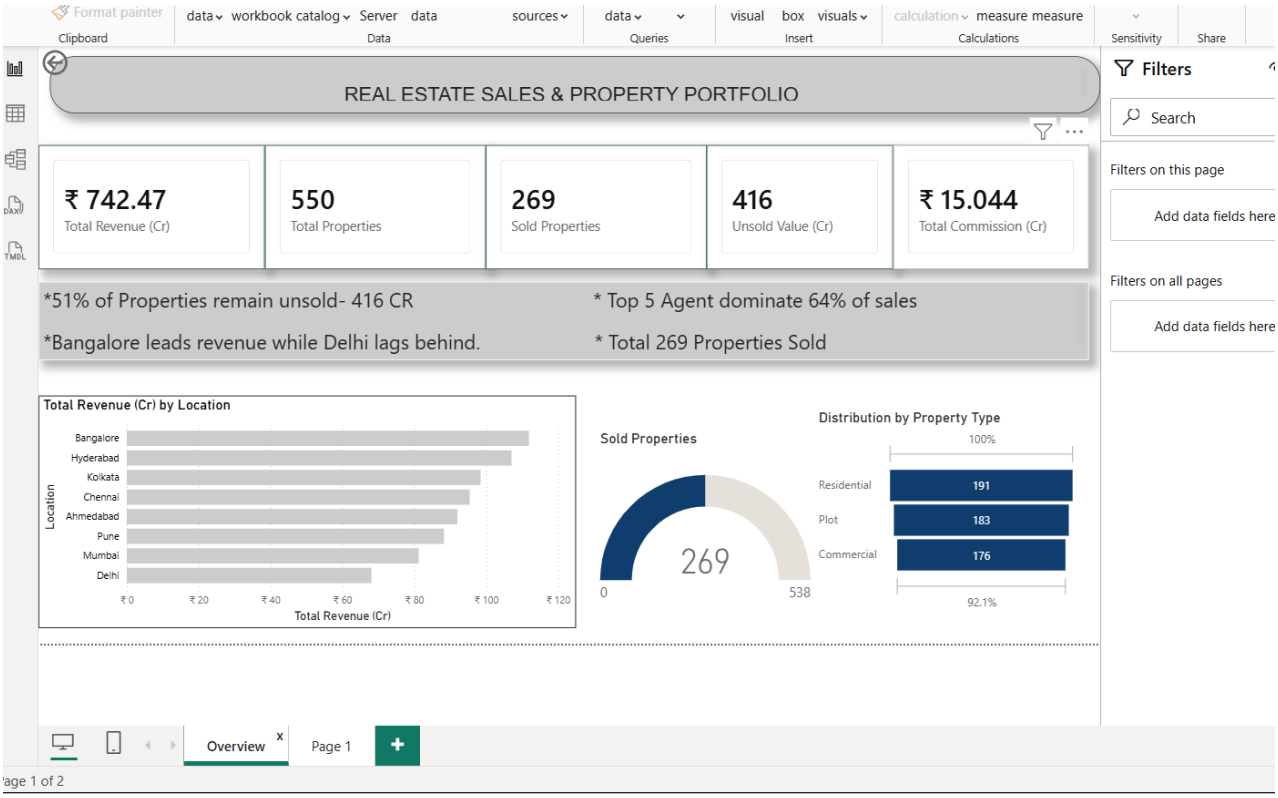
- Total Sales = SUM(Properties[Sale Price])
- Unsold Inventory = SUMX(FILTER(Properties, Properties[Property Status]="Active"), Properties[Current Value])
- Agent Revenue = SUMX(FILTER(Properties, Properties[Agent Name]="Joshua Smith"), Properties[Sale Price])
- Conversion Rate = DIVIDE(CALCULATE(COUNTROWS(Properties), Properties[Property Status]="Sold"), COUNTROWS(Properties))

Appendix C — AI Prompts Used

Prompts with Microsoft Copilot

1. *“Summarize city-wise sales performance and highlight underperforming markets.”*
2. *“Identify top-performing agents and calculate their revenue contribution.”*
3. *“Explain seasonal demand cycles based on monthly sales trends.”*
4. *“Generate a business recommendation to reduce unsold vacant inventory.”*
5. *“Provide a narrative comparing average property prices across cities.”*

Appendix D — Screenshot Gallery



 Code Snippet 1: Example Excel formula (=SUMIF(Location,"Bangalore",SalePrice)) used for city-wise revenue calculation.

 Code Snippet 2: Example DAX measure (Unsold Inventory = SUMX(FILTER(Properties, Properties[Property Status]="Active"), Properties[Current Value])) used for unsold inventory value.